

LAPORAN UJIAN PRAKTIKUM SEMESTER GENAP
BASIS DATA 2

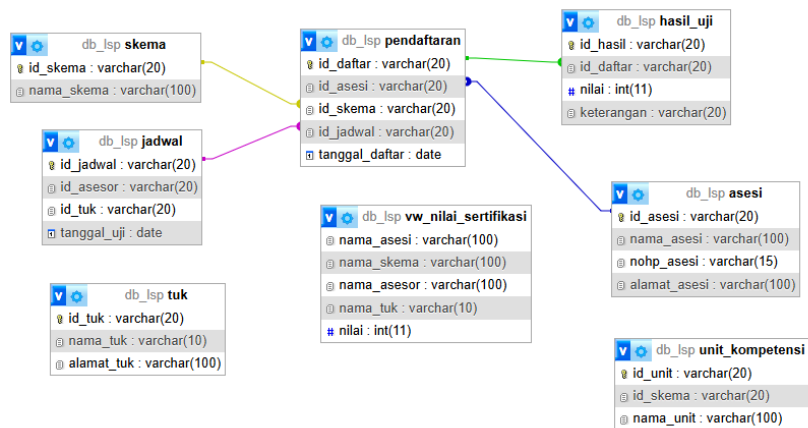


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PROGRAM STUDI DIII TEKNIK INFORMATIKA
POLITEKNIK PURBAYA TEGAL
TAHUN 2025/2026

1. Erd



2. Buat Database Dengan Nama Db_Lsp

```
MariaDB [(none)]> CREATE DATABASE db_lsp;
Query OK, 1 row affected (0.003 sec)
```

3. Buat Tabel Dengan Cli

a. Tabel Asesor

```
MariaDB [(none)]> USE db_lsp;
Database changed
MariaDB [db_lsp]> CREATE TABLE asesor (
  -> id_asesor VARCHAR(20) PRIMARY KEY,
  -> nama_asesor VARCHAR(100),
  -> nohp_asesor VARCHAR(15)
  -> );
Query OK, 0 rows affected (0.022 sec)
```

b. Tabel Asesi

```
MariaDB [db_lsp]> CREATE TABLE asesesi (
  -> id_asesi VARCHAR(20) PRIMARY KEY,
  -> nama_asesi VARCHAR(100),
  -> nohp_asesi VARCHAR(15),
  -> alamat_asesi VARCHAR(100)
  -> );
Query OK, 0 rows affected (0.025 sec)

MariaDB [db_lsp]> DESC asesesi;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id_asesi | varchar(20) | NO | PRI | NULL | |
| nama_asesi | varchar(100) | YES | | NULL | |
| nohp_asesi | varchar(15) | YES | | NULL | |
| alamat_asesi | varchar(100) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.017 sec)
```

c. Tabel Skema

```
MariaDB [db_lsp]> CREATE TABLE skema (  
  -> id_skema VARCHAR(20) PRIMARY KEY,  
  -> nama_skema VARCHAR(100)  
  -> );  
Query OK, 0 rows affected (0.025 sec)  
  
MariaDB [db_lsp]> DESC skema;  
+-----+-----+-----+-----+-----+-----+  
| Field      | Type      | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| id_skema   | varchar(20) | NO   | PRI | NULL    |       |  
| nama_skema | varchar(100) | YES  |     | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
2 rows in set (0.016 sec)
```

d. Tabel Unit_Kompetensi

```
MariaDB [db_lsp]> CREATE TABLE unit_kompetensi (  
  -> id_unit VARCHAR(20) PRIMARY KEY,  
  -> nama_unit VARCHAR(100)  
  -> );  
Query OK, 0 rows affected (0.024 sec)  
  
MariaDB [db_lsp]> DESC unit_kompetensi;  
+-----+-----+-----+-----+-----+-----+  
| Field      | Type      | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| id_unit    | varchar(20) | NO   | PRI | NULL    |       |  
| nama_unit  | varchar(100) | YES  |     | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
2 rows in set (0.016 sec)
```

e. Tabel Tuk

```
MariaDB [db_lsp]> CREATE TABLE tuk (  
  -> id_tuk VARCHAR(20) PRIMARY KEY,  
  -> nama_tuk VARCHAR(10),  
  -> alamat_tuk VARCHAR(100)  
  -> );  
Query OK, 0 rows affected (0.023 sec)  
  
MariaDB [db_lsp]> DESC tuk;  
+-----+-----+-----+-----+-----+-----+  
| Field      | Type      | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| id_tuk     | varchar(20) | NO   | PRI | NULL    |       |  
| nama_tuk   | varchar(10) | YES  |     | NULL    |       |  
| alamat_tuk | varchar(100) | YES  |     | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
3 rows in set (0.007 sec)
```

f. Tabel Jadwal

```
MariaDB [db_lsp]> CREATE TABLE jadwal (  
  -> id_jadwal VARCHAR(20) PRIMARY KEY,  
  -> id_asesor VARCHAR(20),  
  -> id_tuk VARCHAR(20),  
  -> tanggal_uji DATE  
  -> );  
Query OK, 0 rows affected (0.023 sec)  
  
MariaDB [db_lsp]> DESC jadwal;  
+-----+-----+-----+-----+-----+-----+  
| Field      | Type      | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| id_jadwal  | varchar(20) | NO   | PRI | NULL    |       |  
| id_asesor  | varchar(20) | YES  |     | NULL    |       |  
| id_tuk     | varchar(20) | YES  |     | NULL    |       |  
| tanggal_uji | date       | YES  |     | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
4 rows in set (0.015 sec)
```

g. Tabel Pendaftaran

```
MariaDB [db_lsp]> CREATE TABLE pendaftaran (
-> id_daftar VARCHAR(20) PRIMARY KEY,
-> id_asesi VARCHAR(20),
-> id_skema VARCHAR(20),
-> id_jadwal VARCHAR(20),
-> tanggal_daftar DATE,
-> FOREIGN KEY (id_asesi) REFERENCES asesi(id_asesi),
-> FOREIGN KEY (id_skema) REFERENCES skema(id_skema),
-> FOREIGN KEY (id_jadwal) REFERENCES jadwal(id_jadwal)
-> );
Query OK, 0 rows affected (0.035 sec)

MariaDB [db_lsp]> DESC pendaftaran;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id_daftar | varchar(20) | NO | PRI | NULL | |
| id_asesi | varchar(20) | YES | MUL | NULL | |
| id_skema | varchar(20) | YES | MUL | NULL | |
| id_jadwal | varchar(20) | YES | MUL | NULL | |
| tanggal_daftar | date | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.015 sec)
```

h. Tabel Hasil_Uji

```
MariaDB [db_lsp]> CREATE TABLE hasil_uji (
-> id_hasil VARCHAR(20) PRIMARY KEY,
-> id_daftar VARCHAR(20),
-> nilai INT,
-> keterangan VARCHAR(20),
-> FOREIGN KEY (id_daftar) REFERENCES pendaftaran(id_daftar)
-> );
Query OK, 0 rows affected (0.026 sec)

MariaDB [db_lsp]> DESC hasil_uji;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| id_hasil | varchar(20) | NO | PRI | NULL | |
| id_daftar | varchar(20) | YES | MUL | NULL | |
| nilai | int(11) | YES | | NULL | |
| keterangan | varchar(20) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.016 sec)
```

4. Masukkan Data Di Masing-Masing Table

a. Tabel Asesor

```
MariaDB [db_lsp]> SELECT*FROM asesor;
+-----+-----+-----+
| id_asesor | nama_asesor | nohp_asesor |
+-----+-----+-----+
| ASR001 | Aisyah Fitri | 081245347623 |
| ASR002 | Dadang Asep | 081245347234 |
| ASR003 | Muhammad Ibnu | 08192356789 |
| ASR004 | Fatimah | 08192674789 |
| ASR005 | Rizky Maulana | 08192743552 |
+-----+-----+-----+
5 rows in set (0.001 sec)
```

b. Tabel Asesi

```
MariaDB [db_lsp]> SELECT*FROM asesi;
```

id_asesi	nama_asesi	nohp_asesi	alamat_asesi
ASE001	Hanin Khaerunnisa	08234561892	Slawi
ASE002	Amelia Yanuar	08234566752	Dukuhturi
ASE003	Sherra Asyta	08234566342	Tegal Barat
ASE004	Nisrina Nabila	08234566246	Lebaksiu
ASE005	Aulia Rahma	089256432	Tegal
ASE006	Dimas Pratama	0892525122	Slawi
ASE007	Fitri Handayani	082345667113	Adiwerna
ASE008	Rizal Saputra	082345667114	Dukuhturi
ASE009	Nabila Putri	082345667115	Talang
ASE010	Fajar Nugroho	082345667116	Pangkajene
ASE011	Lutfi Hidayat	082345667117	Tegal
ASE012	Rina Oktavia	082345667118	Slawi
ASE013	Yusuf Maulana	082345667119	Balapulang
ASE014	Dewi Lestari	082345667120	Lebaksiu
ASE015	Anisa Putri	082345667121	Tegal
ASE016	Bagus Setiawan	082345667122	Dukuhturi
ASE017	Cahya Ramadhan	082345667123	Talang
ASE018	Dian Pratiwi	082345667124	Pangkajene
ASE019	Eko Saputro	082345667125	Slawi
ASE020	Gita Maharani	082345667126	Adiwerna

```
20 rows in set (0.001 sec)
```

c. Tabel Skema

```
MariaDB [db_lsp]> SELECT*FROM skema;
```

id_skema	nama_skema
SKM001	Web Developer
SKM002	Network Engineer
SKM003	Database Administrator
SKM004	Programmer
SKM005	UI UX Designer

```
5 rows in set (0.001 sec)
```

d. Tabel Unit_Kompetensi

```
MariaDB [db_lsp]> SELECT*FROM unit_kompetensi;
```

id_unit	id_skema	nama_unit
UK001	SKM001	HTML Dasar
UK002	SKM001	CSS Dasar
UK003	SKM001	JavaScript Dasar
UK004	SKM001	PHP Dasar
UK005	SKM002	Konfigurasi Router
UK006	SKM002	Konfigurasi Switch
UK007	SKM002	TCP IP
UK008	SKM002	Troubleshooting Jaringan
UK009	SKM003	DDL Database
UK010	SKM003	DML Database
UK011	SKM003	Database Security
UK012	SKM003	Backup Database
UK013	SKM004	Pemrograman Dasar
UK014	SKM004	OOP
UK015	SKM004	Struktur Data
UK016	SKM004	Debugging Program
UK017	SKM005	User Research
UK018	SKM005	Wireframe
UK019	SKM005	Prototype
UK020	SKM005	Usability Testing

```
20 rows in set (0.001 sec)
```

e. Tabel Tuk

```
MariaDB [db_lsp]> SELECT*FROM tuk;
+-----+-----+-----+
| id_tuk | nama_tuk | alamat_tuk |
+-----+-----+-----+
| TUK001 | TUK Tegal | Tegal      |
| TUK002 | TUK Slawi | Slawi      |
| TUK003 | TUK Brebes | Brebes     |
| TUK004 | TUK Pemala | Pemalang   |
| TUK005 | TUK Pekalo | Pekalongan |
+-----+-----+-----+
5 rows in set (0.001 sec)
```

f. Tabel Jadwal

```
MariaDB [db_lsp]> SELECT*FROM jadwal;
+-----+-----+-----+-----+
| id_jadwal | id_asesor | id_tuk | tanggal_uji |
+-----+-----+-----+-----+
| JDW001    | ASR001    | TUK001 | 2026-07-25   |
| JDW002    | ASR002    | TUK002 | 2026-07-25   |
| JDW003    | ASR003    | TUK003 | 2026-07-26   |
| JDW004    | ASR004    | TUK004 | 2026-07-26   |
| JDW005    | ASR005    | TUK005 | 2026-07-27   |
| JDW006    | ASR001    | TUK002 | 2026-07-27   |
| JDW007    | ASR002    | TUK003 | 2026-07-28   |
| JDW008    | ASR003    | TUK004 | 2026-07-28   |
| JDW009    | ASR004    | TUK005 | 2026-07-29   |
| JDW010    | ASR005    | TUK001 | 2026-07-29   |
+-----+-----+-----+-----+
10 rows in set (0.001 sec)
```

g. Tabel Pendaftaran

```
MariaDB [db_lsp]> SELECT*FROM pendaftaran;
+-----+-----+-----+-----+-----+
| id_daftar | id_asesi | id_skema | id_jadwal | tanggal_daftar |
+-----+-----+-----+-----+-----+
| PDF001    | ASE001   | SKM001   | JDW001    | 2026-07-20      |
| PDF002    | ASE002   | SKM001   | JDW001    | 2026-07-20      |
| PDF003    | ASE003   | SKM002   | JDW002    | 2026-07-20      |
| PDF004    | ASE004   | SKM002   | JDW002    | 2026-07-20      |
| PDF005    | ASE005   | SKM003   | JDW003    | 2026-07-20      |
| PDF006    | ASE006   | SKM003   | JDW003    | 2026-07-20      |
| PDF007    | ASE007   | SKM004   | JDW004    | 2026-07-21      |
| PDF008    | ASE008   | SKM004   | JDW004    | 2026-07-21      |
| PDF009    | ASE009   | SKM005   | JDW005    | 2026-07-21      |
| PDF010    | ASE010   | SKM005   | JDW005    | 2026-07-21      |
| PDF011    | ASE011   | SKM001   | JDW006    | 2026-07-22      |
| PDF012    | ASE012   | SKM002   | JDW007    | 2026-07-22      |
| PDF013    | ASE013   | SKM003   | JDW008    | 2026-07-22      |
| PDF014    | ASE014   | SKM004   | JDW009    | 2026-07-22      |
| PDF015    | ASE015   | SKM005   | JDW010    | 2026-07-22      |
| PDF016    | ASE016   | SKM001   | JDW001    | 2026-07-23      |
| PDF017    | ASE017   | SKM002   | JDW002    | 2026-07-23      |
| PDF018    | ASE018   | SKM003   | JDW003    | 2026-07-23      |
| PDF019    | ASE019   | SKM004   | JDW004    | 2026-07-23      |
| PDF020    | ASE020   | SKM005   | JDW005    | 2026-07-23      |
+-----+-----+-----+-----+-----+
20 rows in set (0.001 sec)
```

h. Tabel Hasil_Uji

```
MariaDB [db_lsp]> SELECT*FROM hasil_uji;
```

id_hasil	id_daftar	nilai	keterangan
HSL001	PDF001	85	Kompeten
HSL002	PDF002	78	Kompeten
HSL003	PDF003	90	Kompeten
HSL004	PDF004	70	Belum Kompeten
HSL005	PDF005	88	Kompeten
HSL006	PDF006	82	Kompeten
HSL007	PDF007	76	Kompeten
HSL008	PDF008	65	Belum Kompeten
HSL009	PDF009	92	Kompeten
HSL010	PDF010	80	Kompeten
HSL011	PDF011	84	Kompeten
HSL012	PDF012	75	Kompeten
HSL013	PDF013	89	Kompeten
HSL014	PDF014	68	Belum Kompeten
HSL015	PDF015	94	Kompeten
HSL016	PDF016	81	Kompeten
HSL017	PDF017	77	Kompeten
HSL018	PDF018	86	Kompeten
HSL019	PDF019	72	Belum Kompeten
HSL020	PDF020	91	Kompeten

20 rows in set (0.001 sec)

5. Buat View

```
MariaDB [db_lsp]> CREATE VIEW vw_nilai_sertifikasi AS SELECT
-> a.nama_asesi,
-> s.nama_skema,
-> asr.nama_asesor,
-> t.nama_tuk,
-> h.nilai
-> FROM hasil_uji h
-> JOIN pendaftaran p ON h.id_daftar = p.id_daftar
-> JOIN asesi a ON p.id_asesi = a.id_asesi
-> JOIN skema s ON p.id_skema = s.id_skema
-> JOIN jadwal j ON p.id_jadwal = j.id_jadwal
-> JOIN asesor asr ON j.id_asesor = asr.id_asesor
-> JOIN tuk t ON j.id_tuk = t.id_tuk;
```

Query OK, 0 rows affected (0.009 sec)

```
MariaDB [db_lsp]> SELECT*FROM vw_nilai_sertifikasi;
```

nama_asesi	nama_skema	nama_asesor	nama_tuk	nilai
Hanin Khaerunnisa	Web Developer	Aisyah Fitri	TUK Tegal	85
Amelia Yanuar	Web Developer	Aisyah Fitri	TUK Tegal	78
Lutfi Hidayat	Web Developer	Aisyah Fitri	TUK Slawi	84
Bagus Setiawan	Web Developer	Aisyah Fitri	TUK Tegal	81
Sherra Asyta	Network Engineer	Dadang Asep	TUK Slawi	90
Nisrina Nabila	Network Engineer	Dadang Asep	TUK Slawi	70
Rina Oktavia	Network Engineer	Dadang Asep	TUK Brebes	75
Cahya Ramadhan	Network Engineer	Dadang Asep	TUK Slawi	77
Aulia Rahma	Database Administrator	Muhammad Ibnu	TUK Brebes	88
Dimas Pratama	Database Administrator	Muhammad Ibnu	TUK Brebes	82
Yusuf Maulana	Database Administrator	Muhammad Ibnu	TUK Pemala	89
Dian Pratiwi	Database Administrator	Muhammad Ibnu	TUK Brebes	86
Fitri Handayani	Programmer	Fatimah	TUK Pemala	76
Rizal Saputra	Programmer	Fatimah	TUK Pemala	65
Dewi Lestari	Programmer	Fatimah	TUK Pekalo	68
Eko Saputro	Programmer	Fatimah	TUK Pemala	72
Nabila Putri	UI UX Designer	Rizky Maulana	TUK Pekalo	92
Fajar Nugroho	UI UX Designer	Rizky Maulana	TUK Pekalo	80
Anisa Putri	UI UX Designer	Rizky Maulana	TUK Tegal	94
Gita Maharani	UI UX Designer	Rizky Maulana	TUK Pekalo	91

20 rows in set (0.004 sec)

6. Subquery

a. Menampilkan Peserta Dengan Nilai Diatas Rata-Rata

```
MariaDB [db_lsp]> SELECT a.nama_asesi, h.nilai
-> FROM asesi a
-> JOIN pendaftaran p ON a.id_asesi = p.id_asesi
-> JOIN hasil_uji h ON p.id_daftar = h.id_daftar
-> WHERE h.nilai >
-> (
-> SELECT AVG(nilai)
-> FROM hasil_uji
-> );
```

nama_asesi	nilai
Hanin Khaerunnisa	85
Sherra Asyta	90
Aulia Rahma	88
Dimas Pratama	82
Nabila Putri	92
Lutfi Hidayat	84
Yusuf Maulana	89
Anisa Putri	94
Dian Pratiwi	86
Gita Maharani	91

10 rows in set (0.002 sec)

- b. Menampilkan Asesor Yang Belum Memiliki Jadwal

```
MariaDB [db_lsp]> SELECT nama_asesor
-> FROM asesor
-> WHERE id_asesor NOT IN
-> (
-> SELECT id_asesor
-> FROM jadwal
-> );
Empty set (0.001 sec)
```

- c. Menampilkan Skema Dengan Peserta Terbanyak

```
MariaDB [db_lsp]> SELECT nama_skema
-> FROM skema
-> WHERE id_skema =
-> (
-> SELECT id_skema
-> FROM pendaftaran
-> GROUP BY id_skema
-> ORDER BY COUNT(*) DESC
-> LIMIT 1
-> );
+-----+
| nama_skema |
+-----+
| Database Administrator |
+-----+
1 row in set (0.001 sec)

MariaDB [db_lsp]>
```